

CLASS 8 · ICSE · SELINA CONCISE MATHEMATICS · SHORT NOTES

# Chapter 18: Constructions

Topic-wise short notes with solved examples

Ruler-and-compass constructions of angles and of quadrilaterals, parallelograms, rectangles, rhombuses and squares from given measurements.

## 1 Basic constructions

- Use only a ruler and compasses unless a protractor is allowed. Keep all arcs visible.
- **Perpendicular bisector of a segment AB:** with centres A and B and the same radius (more than half of AB), draw arcs on both sides. Join the two points where the arcs cross.
- **Bisector of an angle:** with the vertex as centre, cut both arms. From these two points draw equal arcs to cross. Join the vertex to the crossing point.
- **Angle of  $60^\circ$ :** with centre A draw an arc cutting the line at P. With centre P and the same radius, cut the arc at Q. Then  $\angle QAP = 60^\circ$ .

| Angle       | How to construct                                    |
|-------------|-----------------------------------------------------|
| $60^\circ$  | One step of the radius along the arc                |
| $120^\circ$ | Two steps of the radius along the arc               |
| $90^\circ$  | Bisect the angle between $60^\circ$ and $120^\circ$ |
| $30^\circ$  | Bisect $60^\circ$                                   |
| $45^\circ$  | Bisect $90^\circ$                                   |
| $75^\circ$  | Bisect the angle between $60^\circ$ and $90^\circ$  |
| $105^\circ$ | Bisect the angle between $90^\circ$ and $120^\circ$ |
| $135^\circ$ | $90^\circ + 45^\circ$                               |

## 2 Before constructing a quadrilateral

- Draw a **rough sketch** and mark the given measurements.
- A quadrilateral needs **five** independent measurements. Special quadrilaterals need fewer because their properties give the rest.

| Figure        | Measurements needed |
|---------------|---------------------|
| Quadrilateral | 5                   |
| Parallelogram | 3                   |
| Rectangle     | 2                   |
| Rhombus       | 2                   |
| Square        | 1                   |

- Usually one triangle of the figure is constructed first; the fourth vertex is then found with arcs.

### 3 Quadrilaterals

| Given                         | Method                                                                                 |
|-------------------------------|----------------------------------------------------------------------------------------|
| 4 sides and 1 diagonal        | Draw the diagonal; make a triangle on each side of it with the given sides             |
| 4 sides and 1 angle           | Draw two sides with the angle between them; cut arcs for the other two sides           |
| 3 sides and 2 included angles | Draw the middle side; make the angles at its ends; cut off the other two sides         |
| 3 sides and 2 diagonals       | Draw the triangle with one side and the two segments from its ends; complete with arcs |

**Example 3.1** Construct quadrilateral ABCD with  $AB = 5$  cm,  $BC = 4$  cm,  $CD = 4.5$  cm,  $DA = 3.5$  cm and  $BD = 6$  cm.

#### SOLUTION

Draw  $BD = 6$  cm. With centres B and D and radii 5 cm and 3.5 cm, draw arcs meeting at A. On the other side of BD, with centres B and D and radii 4 cm and 4.5 cm, draw arcs meeting at C.

**Join AB, AD, BC and CD**

**Example 3.2** Construct quadrilateral PQRS with  $PQ = 5$  cm,  $QR = 4$  cm,  $RS = 3.5$  cm,  $\angle Q = 60^\circ$  and  $\angle R = 90^\circ$ .

#### SOLUTION

Draw  $QR = 4$  cm. At Q construct  $60^\circ$  and cut off  $QP = 5$  cm. At R construct  $90^\circ$  and cut off  $RS = 3.5$  cm.

**Join PS**

### 4 Parallelograms

- Use: opposite sides are equal and parallel; the diagonals bisect each other.

| Given                                   | Method                                                                                                     |
|-----------------------------------------|------------------------------------------------------------------------------------------------------------|
| 2 adjacent sides and the included angle | Draw the two sides with the angle; complete with arcs equal to the opposite sides                          |
| 2 adjacent sides and a diagonal         | Draw the triangle formed by them; complete with arcs                                                       |
| 2 diagonals and the angle between them  | Draw one diagonal and mark its mid-point O; draw the other through O at the given angle, half on each side |
| 1 side and 2 diagonals                  | Draw the triangle with the side and the two half-diagonals; extend each half-diagonal by its own length    |

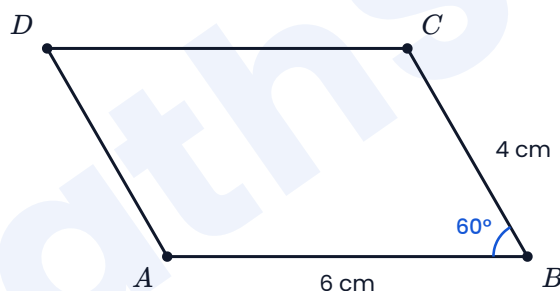
**Example 4.1** Construct parallelogram ABCD with  $AB = 6$  cm,  $BC = 4$  cm and  $\angle B = 60^\circ$ .

**SOLUTION**

Draw  $AB = 6$  cm. At B construct  $60^\circ$  and cut off  $BC = 4$  cm.

With centre A and radius 4 cm, and centre C and radius 6 cm, draw arcs meeting at D.

**Join AD and CD**



Draw AB and BC first, then find D with two arcs

**Example 4.2** Construct a parallelogram with diagonals 8 cm and 6 cm that meet at  $60^\circ$ .

**SOLUTION**

Draw  $AC = 8$  cm and mark its mid-point O. Through O draw a line making  $60^\circ$  with AC.

On this line mark  $OB = OD = 3$  cm on opposite sides of O.

**Join AB, BC, CD and DA**

## 5 Rectangles

| Given                   | Method                                                                                        |
|-------------------------|-----------------------------------------------------------------------------------------------|
| Length and breadth      | Draw the length; construct $90^\circ$ at both ends; cut off the breadth on each               |
| One side and a diagonal | Draw the side AB; construct $90^\circ$ at B; from A cut this line with the diagonal as radius |

**Example 5.1** Construct rectangle ABCD with  $AB = 6$  cm and  $AC = 7.5$  cm.

**SOLUTION**

Draw  $AB = 6$  cm. At B construct  $90^\circ$ . With centre A and radius 7.5 cm, cut the perpendicular at C.

With centre A and radius BC, and centre C and radius 6 cm, draw arcs meeting at D.

**Join AD and CD. (BC will measure 4.5 cm.)**

## 6 Rhombuses

| Given                 | Method                                                                                                |
|-----------------------|-------------------------------------------------------------------------------------------------------|
| Side and one angle    | Draw two sides with the angle; complete with arcs equal to the side                                   |
| Side and one diagonal | Draw the diagonal; from both ends draw arcs equal to the side, on both sides of it                    |
| Two diagonals         | Draw one diagonal; construct its perpendicular bisector; mark half of the other diagonal on each side |

**Example 6.1** Construct a rhombus with diagonals 8 cm and 6 cm.

**SOLUTION**

Draw  $AC = 8$  cm and its perpendicular bisector, meeting AC at O.

Mark  $OB = OD = 3$  cm on the bisector.

**Join AB, BC, CD and DA. Each side will measure 5 cm.**

## 7 Squares

| Given    | Method                                                                                                        |
|----------|---------------------------------------------------------------------------------------------------------------|
| Side     | Draw the side; construct $90^\circ$ at both ends; cut off the side on each; join                              |
| Diagonal | Draw the diagonal; construct its perpendicular bisector; mark half the diagonal on each side of the mid-point |

**Example 7.1** Construct a square with diagonal 6 cm.

**SOLUTION**

Draw  $AC = 6$  cm and its perpendicular bisector through O.

**Mark  $OB = OD = 3$  cm on it and join AB, BC, CD and DA**

## 8 Common mistakes

- Starting without a rough sketch.
- Using a protractor where a compass construction is required.
- Erasing the construction arcs.
- Marking the full diagonal on each side of O instead of half.
- Drawing the second triangle on the same side of the diagonal as the first.
- Not checking that the first triangle is possible (sum of two sides greater than the third).

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